



AI-Driven Exploratory Data Analysis on Alpha Innovations' Marketing Campaigns

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Motivation/Introduction

In today's competitive digital landscape, Alpha Innovations relies on data-driven strategies to enhance marketing effectiveness. This project applies Exploratory Data Analysis (EDA) to uncover trends, patterns, and insights from campaign data. By integrating machine learning, sentiment analysis, and statistical techniques, the goal is to evaluate engagement, predict performance, and optimize campaign outcomes.

SCOPE of the Project

The scope includes analysing a subset of a Marketing Campaign Performance dataset specific to Alpha Innovations, consisting of 40,051 records over a 2-year period. The analysis focuses on campaign types, performance metrics like ROI and engagement, and customer segmentation. Advanced models such as XGBoost, Random Forest, and ARIMA are applied to support prediction and strategy formulation. The project also explores sentiment trends using NLP tools like VADER.

Methodology

[1] Marketing Campaign Performance

[2] Data Preprocessing

- Handle Missing Values
- Standardize dates

[3] Data Visualizations

- Bar
- Heat maps

[6] Model Building & Evaluation

- XG Boost
- Random Forest
- ARIMA
- Naïve Byes

[5] Feature Engineering

- Feature Selection
 - ✓ PCA
- Feature Extraction
 - ✓ Filter Approach

[4] Correlation Analysis

- Haet Maps to identify patterns

[7] Prescriptive Insights

- Recommendations based on model outputs

[8] Final Reporting

- Summarise Insights
- Visualize Findings

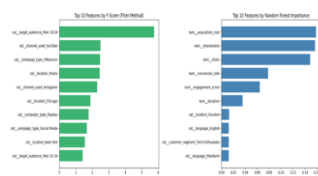


Fig 1: Feature Extraction (Filter method) PCA and filtering were applied for feature extraction. Acquisition_cost was extracted as important feature through filtering method.

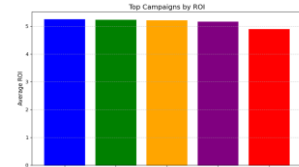


Fig 2: Prescriptive Analytics AI-powered modifications adjust advertising placements, timing, and content in response to changes in emotion.



Fig 3: Measures of Engagement Correlations between various engagement drivers are displayed in a heatmap.

Results

Table 1: Ensemble Models Performance Evaluation

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
Random Forest	85.6%	89.1%	78.3%	83.4%	91.0%
Gradient Boosting	87.2%	84.7%	85.5%	85.1%	93.2%
Voting Classifier	88.0%	88.4%	87.1%	87.1%	94.5%

- Random Forest (SMOTE): Best for precision; reduces false positives.
- Gradient Boosting (SMOTE): Best for recall; captures more positives.
- Voting Classifier (SMOTE): Best overall; balances precision and recall effectively.

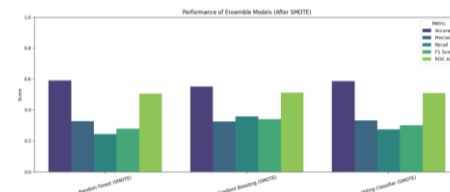


Fig 4: Performance of Ensemble Models

- Gradient Boosting offers the most balanced performance across all metrics, especially in recall and F1 score.
- Voting Classifier shows competitive ROC AUC but underperforms in recall and F1.
- Random Forest shows decent accuracy and precision but weaker recall. Gradient Boosting with SMOTE is the most well-rounded and suitable choice, especially when dealing with imbalanced classes where recall and F1 score are more critical than raw accuracy.

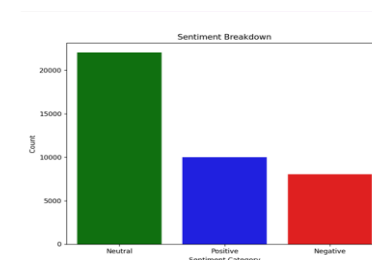


Fig 5: Sentiment Breakdown

- The majority of advertising did not elicit significant emotional reactions, as indicated by the neutral attitude dominating (biggest bar). There is less negative feedback than positive, indicating better campaign performance overall.

Conclusion/ Summary

This project demonstrates the value of combining EDA with AI tools to gain strategic marketing insights. The findings emphasize the significance of customer engagement and sentiment in driving campaign success. XGBoost emerges as the most effective model for sentiment prediction, and the analysis supports smarter budgeting and targeting decisions.

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GITHUB LINK: https://github.com/keerthanaAR25/EDA_PROJECT

Acknowledgement/ References:

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